ADRIATICA D.O.O.

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Marine Engineering & Regulatory Compliance

Sustainable In-Water Cleaning: Engineering Management for European Coastal Yacht Compliance

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Abstract

As IMO MEPC.378(80) Guidelines transition toward structured enforcement convergence across Port State Control (PSC) regimes, biofouling management is shifting from voluntary guidance toward structured operational oversight. While in-water cleaning offers a cost-effective alternative to dry-docking, it requires stringent adherence to environmental standards and technical documentation across different European regions. Adriatica D.O.O. provides engineering management services to plan, coordinate, and certify in-water cleaning operations. Although Adriatica D.O.O. operates primarily in the Adriatic and Mediterranean regions, our compliance framework is aligned with Port State Control inspection expectations under the Paris MoU framework [5, 9], ensuring continued compliance for 25m+ and 50m+ vessels without disruption to operational scheduling.

1. Context & Regulatory Landscape

Under the IMO MEPC.378(80) Guidelines [1], biofouling management is progressing from voluntary guidance toward structured enforcement convergence. European coastal regimes, including the Mediterranean, Adriatic, and Baltic Seas, are intensifying scrutiny of arriving vessels [5]. Regional enforcement intensity varies; for example, the Baltic Sea region is subject to heightened environmental sensitivity under HELCOM coordination frameworks [9]. For motor yachts and sailing vessels (25m+), reliance on reactive cleaning leads to:

- High costs of unscheduled dry-docking.
- Charter schedule disruptions and operational downtime.
- Regulatory non-compliance penalties across different jurisdictions.

2. Methodology: Adriatica Engineering Management Framework

In-water cleaning (cleaning without hauling the vessel) provides a sustainable solution when executed professionally. It offers:

- Cost Efficiency: Cost reductions typically ranging between 60–70% compared to unscheduled dry-docking [8].
- Operational Continuity: Cleaning performed during guest turnaround or downtime.
- Technical Focus: Specialized removal of macro-fouling from niche areas (propellers, thruster tunnels, rudder pintles).

Adriatica D.O.O. manages the entire cleaning process through a structured approach, ensuring quality and compliance. It is important to note that Adriatica D.O.O. does not perform physical cleaning operations; acting as Owner's Technical Representative for biofouling management, we design, manage, and technically certify the process.

- Pre-Cleaning Risk Assessment: Technical evaluation of fouling levels to determine the necessary cleaning intensity based on the vessel's operational history. Technical diagrams identifying all vessel-specific high-risk zones.
- Certified Partner Coordination: Management of certified diving teams equipped with advanced cleaning technologies, including diver-operated or ROV-based systems, with vacuum-assisted and non-abrasive methods for debris capture. This ensures compliance with stringent environmental standards in diverse ports such as Monaco, Tivat, Dubrovnik, and Kiel.
- Environmental Control & Debris Capture Compliance: Ensuring diver technology selection adheres to debris containment and sensitivity regarding discharge in European coastal waters, particularly in ecologically sensitive areas like the Baltic or specific Mediterranean marine reserves. In the Baltic Sea, lower salinity and enclosed sea dynamics increase the risk of invasive species establishment, making debris capture particularly critical under HELCOM coordination. In jurisdictions with strict port authority enforcement (e.g., Monaco), failure to contain cleaning debris may trigger Port State Control or port authority enforcement action, including administrative penalties or temporary operational restrictions, depending on jurisdiction [3].
- Quality Assurance & Reporting: Oversight of the operation, including high-resolution photographic documentation, ensuring the process is logged in the vessel's Biofouling Record Book (BFRB) [6].

3. Case Study: Economic Comparison

Feature	Scenario A: Dry-Docking	Scenario B: Managed In-Water Cleaning
Cost	€30,000	€6,000
Downtime	4 Days	1 Day
Charter Risk	High (Cancellation Likely)	Negligible
Environmental Compliance	Indirect (Yard documentation)	Direct (BFRB + Photographic Evidence)

- Note: Representative mid-season comparison based on European operational averages.
- Conclusion: Managed in-water cleaning provided verified compliance with significant financial and operational advantages. The cleaning result was documented through post-cleaning inspection and recorded in the vessel's Biofouling Record Book (BFRB), establishing an auditable trail for Port State Control.

4. Outcomes & Technical Benefits

Implementing a managed in-water cleaning program delivers measurable operational advantages:

Regulatory Alignment

- Full documentation to satisfy Paris MoU and Class society expectations [5, 6].
- Biofouling Record Book (BFRB) entries with photographic evidence create an audit trail accepted by Port State Control authorities.
- Supports alignment with IMO MEPC.378(80) guidance and anticipated 2026 enforcement convergence.

Operational Continuity

- - Cleaning performed during guest turnaround or scheduled downtime, avoiding charter disruption.
- - Typical cleaning completed within one day, compared to 4–7 days for unscheduled dry-docking.

Cost Savings

- - Up to 80% reduction in direct costs compared to unscheduled dry-docking (€6,000 vs. €30,000 based on European operational averages).
- - Elimination of off-hire penalties and emergency cleaning surcharges (€5,000–€15,000).

Environmental Stewardship

- - Verified debris capture using vacuum-based systems reduces risk of environmental enforcement in sensitive areas such as the Baltic Sea (HELCOM) and Mediterranean marine reserves.
- - Supports the vessel's environmental profile, increasingly a factor in marina access and charter market positioning.

5. Strategic Insight

In-water cleaning, when managed through a structured engineering framework, transforms a reactive maintenance task into a proactive compliance asset. The key is not only the cleaning itself but the documented audit trail that demonstrates responsible biofouling management.

From a regulatory perspective, the transition from IMO voluntary guidance to structured Port State Control enforcement (anticipated by 2026) means that vessels without verifiable biofouling records will face increasing friction. A managed in-water cleaning program provides traceability, risk reduction, and cost certainty. As regional enforcement intensifies toward 2026 across Mediterranean and European coastal waters, vessels with verified cleaning records will experience fewer delays and stronger charter appeal.

6. Conclusion

As 2026 approaches, proactive biofouling governance differentiates compliant operators and protects charter continuity in high-demand Mediterranean seasons. The convergence of IMO MEPC.378(80) guidance with Port State Control enforcement means that documentation, traceability, and niche-area management are becoming prerequisites for seamless Mediterranean entry.

Adriatica D.O.O. offers a structured engineering management solution that bridges regulatory requirements and on-water execution. We invite yacht operators to integrate managed in-water cleaning into their compliance roadmap ahead of 2026 enforcement convergence trends, ensuring operational freedom and regulatory readiness across Mediterranean and European coastal waters.

References

[1] IMO MEPC.378(80): *2023 Guidelines for the Control and Management of Ships' Biofouling to Minimize the Transfer of Invasive Aquatic Species.*

[2] IMO CII Framework: *Technical guidance on the Carbon Intensity Indicator (CII) and its operational efficiency impacts.*

[3] BIMCO / Clean Shipping Alliance: *Industry standards for in-water cleaning and biofouling documentation, emphasizing strict debris capture in sensitive areas.*

[4] IMO PPR Sub-Committee Reports (PPR 13, 2026 Preparatory Work): *Detailing technical measures and enforcement pathways for biofouling management.*

[5] Paris Memorandum of Understanding (Paris MoU) Guidance Notes: *Port State Control procedures and inspection criteria for biofouling documentation and compliance.*

[6] IACS Unified Requirements and relevant classification society rules (RINA, DNV, Bureau Veritas): *Concerning Biofouling Management Plan documentation and compliance in alignment with regional environmental expectations.*

[7] Australian/New Zealand Biofouling Standards (BIOF125): *Mandatory biosecurity models serving as benchmarks for global regulatory alignment.*

[8] Adriatica D.O.O. Engineering Archives: *Internal case studies on superyacht operational efficiency in European & Mediterranean regions.*

[9] Helsinki Commission (HELCOM): *Environmental* guidelines for the Baltic Sea area regarding biofouling and debris containment.

Document Control

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